



Explosive Outbursts in Children With Tourette Syndrome: A Closer Look at Family Functioning Dimensions

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Abstract

Approximately 50% of children with Tourette syndrome (TS) experience explosive outbursts marked by disproportionate aggressive episodes. Although among the most disruptive manifestations of TS, their impact on family functioning remains unclear. This study examined whether explosive outbursts predicted impairments across dimensions of family functioning. A total of 289 parents of children aged 6–14 ($M=10.6$; 75% boys) with TS completed measures assessing tic severity, explosive outbursts, and family functioning. Hierarchical regressions controlling for tics and comorbidities showed that explosive outburst severity significantly predicted impairments in problem solving, role distribution, behavior control, and general functioning, but not in affective responsiveness, affective involvement, or communication. Tics and comorbidities were not significant predictors of any family functioning dimension. Findings suggest that explosive outbursts contribute to organizational and regulatory difficulties, underscoring the importance of targeting these processes in interventions. The absence of associations between explosive outbursts and affective dimensions highlights their distinct clinical profile.

Keywords Tourette syndrome · Tics · Explosive outbursts · Disruptive behavior · Family functioning

Background

Tourette syndrome (TS) is a neurodevelopmental disorder involving persistent motor and phonic tics [1]. Tic expression varies over time and differs in frequency, intensity, and location across individuals [1]. Tics typically begin between ages 4 and 6 as simple motor tics, with phonic tics often emerging one to two years later [28, 37]. Tic severity peaks around ages 10 to 12 and generally decreases over time, though some individuals may continue to experience persistent or even worsening symptoms [26, 42]. The global prevalence of TS is estimated at approximately 0.8% in children and adolescents, with boys diagnosed two to four times

more often than girls [30, 48, 49, 55]. Youth with TS often face psychosocial difficulties, including stigmatization, that can affect their daily lives [13, 43, 45].

In addition to coping with tics, up to 90% of individuals with TS experience comorbid conditions, which can contribute to or exacerbate functional impairments and reduce health-related quality of life [8, 18, 23, 27]. Attention deficit/hyperactivity disorder (ADHD) is the most common comorbidity in childhood and frequently co-occurs with TS. Obsessive-compulsive disorder (OCD), as well as anxiety and disruptive behavior disorders, are also prevalent [14, 23, 27]. Individuals with TS show difficulties in inhibitory control, encompassing challenges in regulating behavior, speech, impulses, and emotional responses [40]. Given this, such deficits in inhibition may contribute to a range of disruptive behaviors. These behaviors appear to fall along a broad continuum of severity, ranging from mild oppositionality to severe dysregulated reactions.

At the end of this continuum are explosive outbursts, which are marked by sudden, disproportionate verbal or physical aggression relative to the trigger, accompanied by a sense of loss of control [2, 7]. Their intensity and unpredictability distinguish them from typical temper tantrums. For example, a child may scream and throw objects in

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response to a minor disruption in routine [6]. Studies report that around 50% of individuals with TS experience explosive outbursts [12, 47]. These outbursts are more common in children and are frequently associated with comorbid ADHD and OCD [7, 10, 41]. Deficits in inhibitory control inherent to TS, combined with impulsivity associated with ADHD and cognitive rigidity associated with OCD, may predispose children with TS to explosive outbursts. A low frustration tolerance is a greater underlying cause of explosive outbursts than deficits in social or emotional skills [32, 33]. When present in children, explosive outbursts are often perceived by parents as one of the most disruptive aspects of the syndrome [17, 24]. Parents of children with TS and aggressive behaviors frequently question their parenting abilities and express uncertainty about how to best support their child [16]. They also report a need for targeted interventions to help their children regulate aggression [22]. Because these behaviors occur primarily at home and require parental management [2], they may have important implications within the family.

Few studies to date have explored the impact of TS on family functioning [3, 46, 53]. Family functioning can be understood as the processes through which families carry out the tasks necessary to support the physical, psychological, and social development of their members [15]. Studies suggest that having a child with TS can disrupt family functioning [19, 34, 46]. Children with TS engage less in family activities, communicate less with adult family members, and experience more familial conflict compared to peers without a diagnosis [19]. While tic severity influences family functioning, comorbid symptoms, particularly those of ADHD and OCD, appear to have a greater impact [11, 19, 34, 53, 54]. Other studies have also examined the role of disruptive behaviors on family functioning. Sukhodolsky et al., [52] found that disruptive aggressive behaviors predicted family dysfunction beyond TS and ADHD diagnostic status. This dysfunction was reflected in family members' expression of anger and disagreement, as well as in levels of commitment, help, and support. Converging evidence also comes from intervention research, with evidence that a therapeutic intervention for explosive outbursts in children may lead to improvements in family functioning, particularly in the quality of relationships between family members and in family involvement [36]. Taken together, these findings suggest that disruptive aggressive behaviors may be associated with difficulties in emotional expression and involvement within families.

Theoretical Framework

The McMaster Model offers a relevant framework to understand family functioning and how it is impacted by explosive

outbursts in families of children with TS. This model conceptualizes family functioning across six dimensions [21]. These dimensions, validated in clinical, educational, and research settings, are not intended to represent an exhaustive list of components of family functioning, but rather dimensions particularly useful in clinical contexts: (1) Problem solving: the family's ability to resolve issues threatening its integrity and functioning; (2) Communication: the clarity and directness of information exchange among members; (3) Role distribution: the clarity and equitability of the established behavior patterns through which family members fulfill essential family functions (e.g., preparing meals, comforting a child); (4) Affective responsiveness: the ability to respond to stimuli with appropriate emotional quality and intensity; (5) Affective involvement: the degree of interest and value placed on each member's interests; (6) Behavior control: the behavioral standards and limits a family sets, and how it regulates behavior across situations involving safety, psychobiological needs, and socialization [21, 38]. This model allows to gain insight into how specific aspects of family functioning can be disrupted by different symptoms. Despite evidence that TS and comorbid symptoms can disrupt family functioning, the specific factors underlying these difficulties remain unclear. In particular, explosive outbursts, reported by parents as especially disruptive, have not been directly examined in relation to specific dimensions of family functioning. The McMaster Model therefore provides a relevant framework to clarify which aspects of family functioning are affected by explosive outbursts and to guide targeted, family-centered interventions.

Objectives

The aim of this study was to assess whether explosive outburst severity predicts impairment in family functioning across the six McMaster dimensions (problem solving, communication, role distribution, affective responsiveness, affective involvement, behavior control) and general functioning. Greater severity of explosive outbursts was expected to be associated with greater impairment in affective responsiveness, affective involvement, and general functioning.

Methods

Participants

Parents of children diagnosed with TS were recruited for this study. Inclusion criteria were: (1) having a child between the ages of 6 and 14 with a diagnosis of TS (as reported by the parent); (2) residing with the child on at least a shared

custody basis; (3) being able to read and understand either French or English. The presence of comorbid conditions was not considered an exclusion criterion. All participants provided written informed consent, and the study was approved by the institutional ethics committee of the Université du Québec à Montréal (project #2024–5786).

Procedure

Participants were recruited internationally through social media networks between January 2024 and October 2025, primarily via the platforms of the Association Québécoise de la Tourette, the Association Française du Syndrome de Gilles de la Tourette, and Tourettes Action. This study was part of a larger research project examining the developmental course of explosive outbursts in children with TS. Parents accessed a single Qualtrics link where they first reviewed the study information and provided informed consent, after which they completed a series of validated questionnaires. The median time to complete the full questionnaire was 42 min. For the present study, instruments assessing sociodemographic information, child's tic severity, explosive outbursts and family functioning were used. All questionnaires were available in French and English. Data cleaning excluded entries that were automated or suspicious, duplicate submissions, and surveys completed in less than 15 min. Participants who did not complete the entire survey, but provided all required measures for this study, were retained if their response time was proportionate. Responses failing at least half of the attention checks or containing more than 10% "I prefer not to answer" items were not included in analyses.

Measures

Sociodemographic and clinical information were collected using a custom questionnaire. It included items to verify inclusion criteria and gather data on parent and child demographics, household income, diagnosis, medication, presence of explosive outbursts according to a specific definition, comorbid conditions, and family structure.

The severity of the child's tics was assessed with the Parent Tic Questionnaire (PTQ) [9, 44], a validated parent-report measure. The PTQ evaluates the presence, frequency, and intensity of 14 common motor tics and 14 vocal tics, with an open field for additional tics. Frequency and intensity are each rated on a 4-point Likert scale, yielding a combined score for each tic (0–8) and a total severity score ranging from 0 to 224 (motor: 0–112; vocal: 0–112).

Explosive outburst severity was measured using the Rage Attack Questionnaire–Revised (RAQ-R) [41], a validated instrument originally developed as a self-report measure

and adapted for parent-report in this study. The RAQ-R comprises 22 items rated on a 4-point Likert scale (0–3), yielding a total severity score ranging from 0 to 66. It is the only available measure providing a severity index for explosive outbursts in TS.

Family functioning was evaluated with the Family Assessment Device (FAD) [4, 20, 50], a validated self-report questionnaire based on the McMaster Model. The FAD comprises 60 items rated on a 4-point Likert scale (1–4) and provides a general score of family functioning, as well as six scores reflecting the model's dimensions: problem solving, communication, roles, affective responsiveness, affective involvement, and behavior control.

Statistical Analyses

All analyses were conducted using SPSS version 30.0 (macOS). Descriptive statistics were computed for sociodemographic variables. Prior to hypothesis testing, data were screened for normality and outliers; assumptions of normality were met, and no extreme values required removal or winsorization.

Primary Analyses

Hierarchical multiple regression analyses were conducted to assess if explosive outburst severity predicted impairment in family functioning dimensions. The analyses included the six McMaster Model dimensions as well as general functioning. Two models were tested for each variable. Tic severity and the presence of comorbid conditions were entered as control variables in Model 1 to account for their established associations with family functioning in families of children with TS [11, 19, 34]. Model 2 added explosive outburst severity to evaluate its contribution to the prediction of family functioning impairment.

Supplementary Analyses

To align with previous studies [11], we first treated comorbid conditions as "any co-occurring condition", which includes several conditions and introduces substantial heterogeneity. To assess robustness, we re-estimated hierarchical multiple regressions using a restricted comorbidity set limited to ADHD and OCD. These two conditions commonly co-occur with TS and have been most consistently linked to family functioning in prior studies of young people with TS [19, 34].

Results

A total of 733 entries were recorded in Qualtrics, of which 550 met the inclusion criteria. After excluding invalid responses, 289 participants provided sufficient data to address the study objectives. The participant flow diagram is shown in Fig. 1.

Descriptive Analyses

Parent characteristics are presented in Table 1. Parents had a mean age of 40.7 years ($SD=5.6$), and 95% identified as women. Most resided in Canada (52%), France (21%), or the United Kingdom (16%). The majority were married/partnered with the child's other parent (71%).

Child characteristics are presented in Table 2. Children had a mean age of 10.6 years ($SD=2.2$), and 76% identified as boys. Explosive outbursts were reported in 87%. Most (94%) had at least one comorbid diagnosis, most commonly ADHD (69%) and affective problems (61%).

Primary Analyses

Problem Solving

Model 1, which included the control variables, accounted for 0.7% of the variance in problem solving ($R^2 = 0.007$) and was not significant, $F(2, 286)=1.01$, $p = .364$. Adding explosive outburst severity in Model 2 significantly increased the explained variance ($\Delta R^2 = 0.036$, $p = .001$).

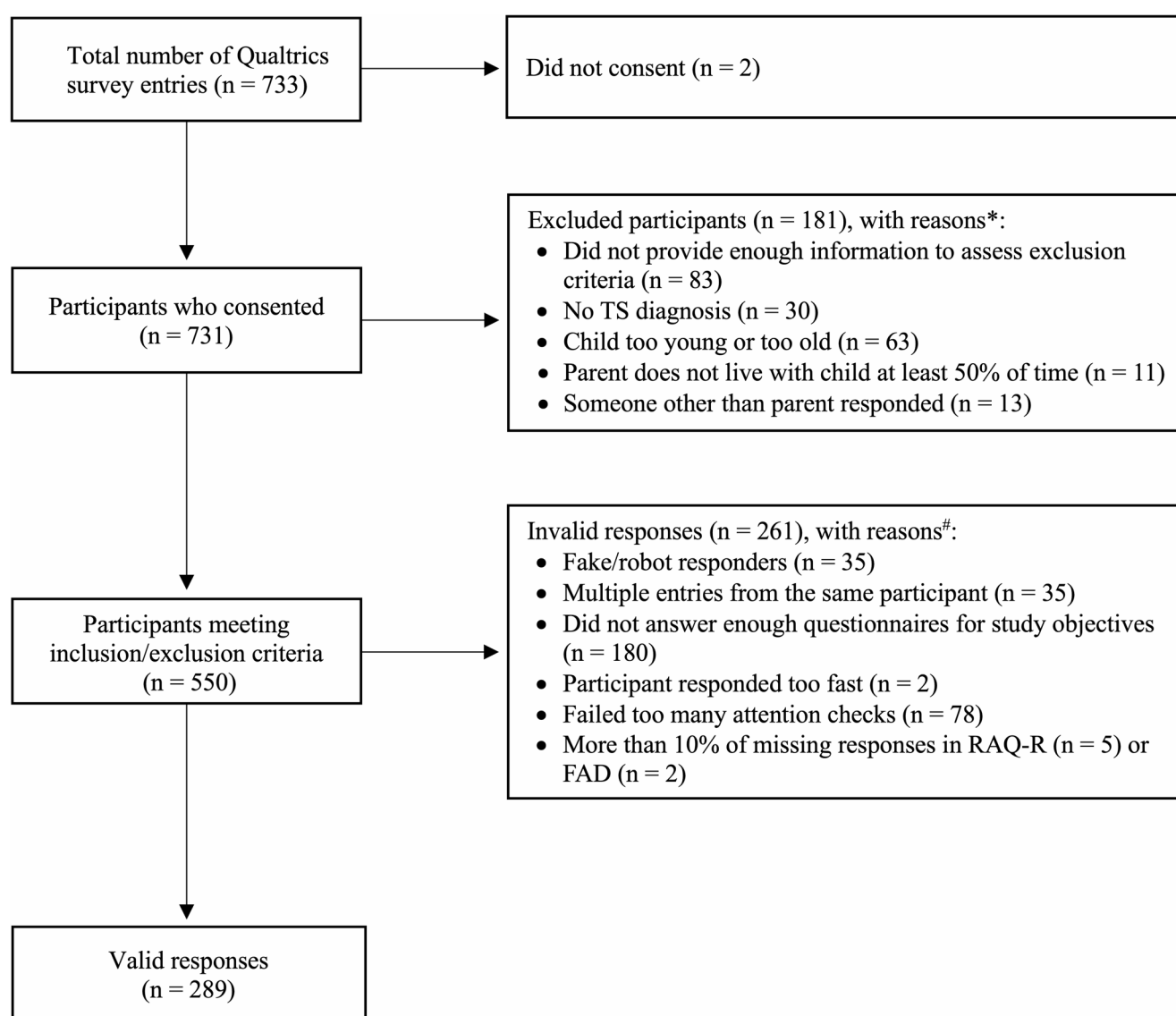


Fig. 1 Participant flow diagram. *Some participants fulfilled multiple exclusion criteria. #Some participants were excluded from the analyses for multiple reasons

Table 1 Demographic characteristics of study participants (parents; $n=289$)

Age in years, mean (SD)	40.7 (5.6)
Sex, number (%)	
Males	20 (6.9%)
Females	269 (93.1%)
Gender, number (%)	
Men	13 (4.5%)
Women	274 (94.8%)
Non-binary	2 (0.7%)
Ethnicity, number (%)	
Caucasian/European	266 (92.0%)
Latin American	5 (1.7%)
East Asian	3 (1.0%)
South Asian	1 (0.3%)
Native/First Nation	2 (0.7%)
Other ¹	9 (3.1%)
Prefer not to answer	3 (1.0%)
Country of residence, number (%)	
Canada	151 (52.2%)
France	60 (20.8%)
United Kingdom	46 (15.9%)
United States of America	20 (6.9%)
Other ²	12 (4.2%)
Parental relationship status, number (%)	
Married/Partnered	205 (70.9%)
Divorced/Separated	72 (24.9%)
Single	9 (3.1%)
Widowed/Partner Deceased	1 (0.3%)
Prefer not to answer	2 (0.7%)
Child's living arrangement if parents are not married/partnered, number (%)	
With mother	38 (13.1%)
With mother and her partner	18 (6.2%)
With father	2 (0.7%)
With father and his partner	0 (0.0%)
Shared custody	24 (8.3%)

SD: standard deviation

¹ Other ethnicity include Jewish, North African, mixed, Western² Other countries include Australia, Belgium, Brazil, Congo, Ireland, Poland, Spain, Sweden, Venezuela

The final model accounted for 4.3% of the variance ($R^2 = 0.043$) and was significant overall, $F(3, 285) = 4.26$, $p = .006$. Explosive outburst severity was a significant positive predictor of impairment in problem solving ($\beta = 0.19$, $p = .001$, 95% CI [0.08, 0.31]), whereas tic severity ($\beta = -0.09$, $p = .124$, 95% CI [-0.21, 0.02]) and presence of comorbid conditions ($\beta = 0.04$, $p = .528$, 95% CI [-0.08, 0.15]) were not significant (see Table 3).

Communication

Model 1 (control variables) explained a small proportion of the variance in communication ($R^2 = 0.007$) and was not

Table 2 Demographic and clinical characteristics of children

Age in years, mean (SD)	10.6 (2.2)
Sex, number (%)	
Males	217 (75.1%)
Females	72 (24.9%)
Gender, number (%)	
Boys	220 (76.1%)
Girls	66 (22.8%)
Non-binary	2 (0.7%)
Other ¹	1 (0.3%)
Explosive outbursts, number (%)	
Presence	252 (87.2%)
Absence	35 (12.1%)
Prefer not to answer	2 (0.7%)
Comorbid diagnoses, number (%) [*]	
ADHD	200 (69.2%)
Affective problems (e.g. anxiety, depression)	177 (61.2%)
Sleep disorders	84 (29.1%)
OCD	81 (28.0%)
Learning disorders	70 (24.2%)
Developmental coordination disorder	36 (12.5%)
Autism	22 (7.6%)
Oppositional defiant disorder	10 (3.5%)
Neurological disorder (e.g. epilepsy)	7 (2.4%)
Sensory processing disorder	7 (2.4%)
Language disorder	4 (1.4%)
Other ²	49 (17.0%)
Siblings, number (%)	
Has siblings	240 (83%)
Only child	49 (17.0%)

ADHD: attention deficit hyperactivity disorder, OCD: obsessive-compulsive disorder

SD: standard deviation

¹ Other gender include questioning gender identity² Other comorbid diagnosis include pending evaluations, subclinical traits without formal diagnosis, giftedness, hypersensitivity, Pediatric Autoimmune Neuropsychiatric Disorders Associated with Streptococcal infections (PANDAS), post-traumatic stress disorder (PTSD), executive dysfunction, cyclical vomiting syndrome, migraines, gastrointestinal disorders, attachment disorder, selective mutism^{*}Several children were reported to have more than one comorbid diagnosis

significant, $F(2, 286) = 1.05$, $p = .350$. Adding explosive outburst severity in Model 2 did not significantly improve model fit ($\Delta R^2 = 0.009$, $p = .099$). The full model was not significant overall, $F(3, 285) = 1.62$, $p = .185$, with $R^2 = 0.017$. No predictor reached significance: explosive outburst severity ($\beta = 0.10$, $p = .099$, 95% CI [-0.02, 0.22]), tic severity ($\beta = 0.07$, $p = .271$, 95% CI [-0.05, 0.18]), and presence of comorbid conditions ($\beta = 0.01$, $p = .897$, 95% CI [-0.11, 0.12]).

Table 3 Hierarchical multiple regression with explosive outburst severity predicting problem solving, controlling for tic severity and presence of comorbid conditions

Predictor	B	SE B	β	t	p	95% CI for B	95% CI for β
Model 1							
Tic severity	-0.01	0.00	-0.06	-1.03	0.303	-0.01, 0.00	-0.18, 0.06
Comorbid conditions	0.71	0.66	0.06	1.06	0.289	-0.60, 2.01	-0.05, 0.18
Model 2							
Tic severity	-0.01	0.00	-0.09	-1.54	0.124	-0.02, 0.00	-0.21, 0.02
Comorbid conditions	0.42	0.66	0.04	0.63	0.528	-0.88, 1.72	-0.08, 0.15
Explosive outburst severity	0.03	0.01	0.19	3.27	0.001***	0.01, 0.05	0.08, 0.31

$R^2 = 0.007$ for Model 1; $R^2 = 0.043$, Significant F $\Delta = 0.001$ for Model 2; *** $p \leq .001$

Table 4 Hierarchical multiple regression with explosive outburst severity predicting role distribution, controlling for tic severity and presence of comorbid conditions

Predictor	B	SE B	β	t	p	95% CI for B	95% CI for β
Model 1							
Tic severity	0.01	0.01	0.05	0.88	0.377	-0.01, 0.02	-0.06, 0.17
Comorbid conditions	2.10	1.14	0.11	1.85	0.065	-0.13, 4.34	-0.01, 0.22
Model 2							
Tic severity	0.00	0.01	0.02	0.28	0.781	-0.01, 0.02	-0.10, 0.13
Comorbid conditions	1.51	1.12	0.08	1.35	0.178	-0.69, 3.71	-0.04, 0.19
Explosive outburst severity	0.07	0.02	0.23	3.96	<0.001***	0.04, 0.11	0.12, 0.35

$R^2 = 0.015$ for Model 1; $R^2 = 0.067$, Significant F $\Delta < 0.001$ for Model 2; *** $p \leq .001$

Role Distribution

With control variables in Model 1, the model accounted for 1.5% of the variance in role distribution ($R^2 = 0.015$), and was not significant, $F(2, 286) = 2.25$, $p = .107$. Adding explosive outburst severity in Model 2 produced a significant improvement ($\Delta R^2 = 0.051$, $p < .001$). The final model accounted for 6.7% of the variance ($R^2 = 0.067$) and was significant overall, $F(3, 285) = 6.80$, $p < .001$. Explosive outburst severity was a positive predictor of impairment in role distribution ($\beta = 0.23$, $p < .001$, 95% CI [0.12, 0.35]), whereas tic severity ($\beta = 0.02$, $p = .781$, 95% CI [-0.10, 0.13]), and presence of comorbid conditions ($\beta = 0.08$, $p = .178$, 95% CI [-0.04, 0.19]) were not significant (see Table 4).

Affective Responsiveness

Model 1 (control variables) accounted for a small proportion of the variance in affective responsiveness ($R^2 = 0.011$), and was not significant, $F(2, 286) = 1.59$, $p = .205$. Adding explosive outburst severity in Model 2 did not improve model fit ($\Delta R^2 = 0.001$, $p = .590$). The final model was not significant, $F(3, 285) = 1.16$, $p = .326$, with $R^2 = 0.012$. Explosive outburst severity was not a significant predictor ($\beta = -0.03$, $p = .590$, 95% CI [-0.15, 0.09]), nor were tic severity ($\beta = 0.11$, $p = .072$, 95% CI [-0.01, 0.23]) or presence of comorbid conditions ($\beta = 0.02$, $p = .752$, 95% CI [-0.10, 0.14]).

Affective Involvement

Model 1 (control variables) explained a small proportion of the variance in affective involvement ($R^2 = 0.007$), and was not significant, $F(2, 286) = 1.05$, $p = .350$. Adding explosive outburst severity in Model 2 did not significantly improve fit ($\Delta R^2 = 0.008$, $p = .121$). The final model was not significant, $F(3, 285) = 1.51$, $p = .212$, with $R^2 = 0.016$. Explosive outburst severity was not a significant predictor ($\beta = 0.09$, $p = .121$, 95% CI [-0.02, 0.21]), nor were tic severity ($\beta = 0.05$, $p = .375$, 95% CI [-0.06, 0.17]) or presence of comorbid conditions ($\beta = 0.04$, $p = .556$, 95% CI [-0.08, 0.15]).

Behavior Control

Model 1, including control variables, accounted for a negligible portion of the variance in behavior control ($R^2 = 0.002$), and was not significant, $F(2, 286) = 0.23$, $p = .796$. Adding explosive outburst severity in Model 2 significantly improved model fit ($\Delta R^2 = 0.014$, $p = .046$). Model 2 was not significant overall, $F(3, 285) = 1.50$, $p = .216$, with $R^2 = 0.015$. Explosive outburst severity was, however, a significant predictor of impairment in behavior control ($\beta = 0.12$, $p = .046$, 95% CI [0.002, 0.24]), whereas tic severity ($\beta = 0.00$, $p = .949$, 95% CI [-0.11, 0.12]) and presence of comorbid conditions ($\beta = 0.02$, $p = .789$, 95% CI [-0.10, 0.13]) were not significant predictors (see Table 5).

Table 5 Hierarchical multiple regression with explosive outburst severity predicting behavior control, controlling for tic severity and presence of comorbid conditions

Predictor	B	SE B	β	t	p	95% CI for B	95% CI for β
Model 1							
Tic severity	0.00	0.01	0.02	0.38	0.705	−0.01, 0.02	−0.09, 0.14
Comorbid conditions	0.47	0.89	0.03	0.53	0.599	−1.28, 2.21	−0.09, 0.15
Model 2							
Tic severity	0.00	0.01	0.00	0.06	0.949	−0.01, 0.01	−0.11, 0.12
Comorbid conditions	0.23	0.89	0.02	0.26	0.798	−1.52, 1.98	−0.10, 0.13
Explosive outburst severity	0.03	0.01	0.12	2.01	0.046*	0.001, 0.06	0.002, 0.24

$R^2 = 0.002$ for Model 1; $R^2 = 0.015$, Significant $F \Delta = 0.046$ for Model 2; * $p < .050$

Table 6 Hierarchical multiple regression with explosive outburst severity predicting general functioning, controlling for tic severity and presence of comorbid conditions

Predictor	B	SE B	β	t	p	95% CI for B	95% CI for β
Model 1							
Tic severity	0.00	0.01	−0.01	−0.22	0.830	−0.02, 0.02	−0.13, 0.10
Comorbid conditions	2.66	1.42	0.11	1.88	0.062	−0.13, 5.45	−0.01, 0.23
Model 2							
Tic severity	−0.01	0.01	−0.05	−0.80	0.423	−0.03, 0.01	−0.16, 0.07
Comorbid conditions	1.95	1.40	0.08	1.40	0.163	−0.80, 4.71	−0.03, 0.20
Explosive outburst severity	0.08	0.02	0.22	3.75	<0.001***	0.04, 0.13	0.11, 0.34

$R^2 = 0.012$ for Model 1; $R^2 = 0.059$ Significant $F \Delta < 0.001$ for Model 2; *** $p \leq .001$

General Functioning

Model 1 (control variables) accounted for 1.2% of the variance in general functioning ($R^2 = 0.012$), and was not significant, $F(2, 286) = 1.76$, $p = .174$. Adding explosive outburst severity in Model 2 significantly improved model fit ($\Delta R^2 = 0.047$, $p < .001$). Model 2 accounted for 5.9% of variance ($R^2 = 0.059$), and was significant overall, $F(3, 285) = 5.93$, $p < .001$. Explosive outburst severity was a significant positive predictor of impairment in general functioning ($\beta = 0.22$, $p < .001$, 95% CI [0.11, 0.34]), whereas tic severity ($\beta = -0.05$, $p = .423$, 95% CI [−0.16, 0.07]) and presence of comorbid conditions ($\beta = 0.08$, $p = .163$, 95% CI [−0.03, 0.20]) were not significant (see Table 6).

Supplementary Analyses

To ensure that our findings were not dependent on how comorbidity was operationalized, we conducted supplementary analyses restricting comorbid conditions to ADHD and OCD, two common and clinically relevant co-occurring disorders in TS. Results were identical with respect to the dimensions affected: the same domains reached (or failed to reach) significance, and the direction and magnitude of standardized coefficients, especially for explosive outburst severity, were substantively unchanged. Model-level indices (R^2 , ΔR^2 , F , p) mirrored the primary analyses without differences of practical relevance. These findings indicate that our conclusions are robust to the operationalization of

comorbidity, whether using a broad composite of conditions or a restricted ADHD/OCD indicator.

Discussion

The present study aimed to examine whether explosive outburst severity in children with TS predicts impairment across distinct dimensions of family functioning, using the McMaster Model as a guiding framework. We first examined the family dimensions affected by explosive outbursts. Our findings indicate that, as expected, greater explosive outburst severity is associated with impairments in general family functioning, with effects in the small-to-moderate range. General functioning, as measured by the FAD, reflects overall satisfaction with family functioning [20, 35]. This result supports the idea that explosive outbursts have a broad impact on family life. Rather than disrupting isolated processes, these outbursts appear to affect multiple aspects of family functioning, contributing to a global sense of impairment. This finding aligns with previous research indicating that parents consider explosive outbursts to be the manifestation of TS that has the greatest impact on family life [24].

Our results further indicate that greater explosive outburst severity predicts impairments in problem solving and role distribution, with effects in the small-to-moderate range. Together, these dimensions reflect core organizational processes within the family system. Problem solving involves a structured process including problem

identification, communication, development of alternative action plans, implementation, and evaluation [21]. Recurrent explosive outbursts may disrupt this process at multiple stages, as families must respond to outbursts combining instrumental demands (e.g., managing daily routines) and affective challenges (e.g., intense emotional reactions). Rather than engaging in deliberate problem identification and alternative planning, families may instead focus on explosive outburst prevention or immediate de-escalation. Over time, this may limit families' capacity to resolve problems in ways that support overall functioning. With regard to role distribution, it seems possible that parents and siblings may adjust their behaviors to anticipate or avoid outbursts, leading to modifications in how responsibilities are assumed within the family. For example, one parent may assume increased responsibility for monitoring daily routine, while the other adapts by avoiding potentially triggering situations. Although such adaptations may seem effective in the short-term, they may also contribute to role overload, blurred role boundaries, or reduced accountability, undermining the clarity and stability of role distribution. This interpretation is consistent with evidence showing that family accommodation in children with TS is driven mainly by aggressive behavior rather than tic severity, and includes modifying the family routine or doing things for the family that are usually the child's responsibility [51].

Our findings also reveal that greater explosive outburst severity predicts impairments in behavior control, with effects in the small range. Effective behavior control involves establishing clear standards for acceptable behavior, combined with an appropriate degree of flexibility to adapt these standards across situations [21]. In the context of explosive outbursts, maintaining consistent and flexible behavior control may be particularly challenging. Families may oscillate between overly rigid responses to prevent outbursts and permissive or inconsistent responses to avoid escalation. This may contribute to a chaotic style of behavior where family members are uncertain about which acceptable behavior apply and how much latitude is allowed. These difficulties may be exacerbated by the child's behavioral inflexibility, reflecting a tendency toward rigid behavior that limits adaptation to changing situational demands [31]. Behavioral inflexibility is hypothesized to be associated with explosive outbursts in TS [39]. Impairments in behavior control may therefore reflect and interact with the child's rigid responding, increasing challenges in establishing and maintaining clear standards while remaining flexible across different situations.

In contrast to the dimensions discussed above, three dimensions of family functioning appear unaffected by explosive outbursts. Contrary to hypotheses, affective responsiveness and affective involvement were not

significantly affected by explosive outburst severity. These findings suggest that, despite recurrent outbursts, families may preserve their capacity to express emotions appropriately and remain emotionally engaged. This contrast with the findings of Sukhodolsky et al., [52], who reported that disruptive aggressive behaviors in children were associated with impairments in these affective dimensions. However, there are fundamental differences between disruptive/oppositional behaviors and explosive outbursts. Unlike oppositional behaviors, explosive outbursts are primarily driven by a lack of control rather than by deliberate aggression or attempts to obtain secondary gain [2, 7, 25, 32]. Children who have explosive outbursts do not intend to provoke their parent or violate rules, as observed in oppositional behaviors. Explosive outbursts in TS have been described as uncharacteristic of children's usual personality [5], in contrast to the more persistent interpersonal tension observed in oppositional defiant disorder. Research on parental attributions suggests that parents' interpretations of children's behavior, particularly regarding intentionality, controllability, and stability, shape their emotional responses and parenting reactions [29]. Consequently, if explosive outbursts are perceived as TS-related behaviors rather than deliberate acts of defiance, it may help preserve parental affective responsiveness and involvement. Our findings also contrast with those of Blanchet [36], who reported post-intervention improvements in affective responsiveness and affective involvement following an intervention for explosive outbursts, suggesting that these dimensions may improve as outbursts decrease. It seems possible that affective responsiveness and involvement may be unrelated to explosive outbursts, but could have changed because of treatment-induced changes in parental capacities, including the acquisition of effective management strategies and increased parental self-efficacy [36]. Our findings also show that communication is not affected by the severity of explosive outburst, suggesting that these outbursts does not affect the clarity and directness of information exchange among family members. This may reflect families' ability to distinguish between explosive outbursts and everyday interactions, thereby maintaining communication patterns outside of these outbursts.

Taken together, our findings suggest that, although comorbidities such as ADHD or OCD are generally thought to exacerbate impairments in family functioning [11, 19, 34], neither tic severity nor the presence of comorbid conditions predicted any family functioning dimensions in the present study. These results indicate that explosive outbursts may have a unique impact beyond these more commonly examined factors when considering family functioning. This does not rule out the relevance of comorbid conditions such as ADHD or OCD for understanding the etiology of explosive outbursts or for guiding interventions, but

suggests that the presence of comorbidities alone provides limited information about the psychosocial consequences experienced by families.

Clinical Implications

From a clinical perspective, our findings highlight several implications for interventions targeting explosive outbursts in young people with TS. The dimensions of family functioning associated with explosive outburst severity, problem solving, role distribution and behavior control, represent processes that may be particularly well suited to brief or group-based interventions. These dimensions involve observable, concrete family practices, such as how problems are addressed, responsibilities are distributed, and behavioral expectations are established. These processes can be addressed through psychoeducation and parental skills-based interventions aimed at enhancing organization, consistency, and flexibility within the family system.

Prior work on interventions for explosive outbursts reported significant decreases in outburst frequency, while intensity showed no measurable change [33]. Yet qualitative findings from the same study indicate that parents nonetheless perceived improvements in outburst intensity, along with positive repercussions on the family [33]. These qualitative findings suggest that clinical improvement may occur even when objective indicators of intensity remain unchanged, consistent with evidence showing that improvements in family functioning may be driven by enhanced parental strategies and increased parental self-efficacy rather than by changes in explosive outburst intensity [36]. Targeting problem solving, role distribution, and behavior control may, in this case, help reduce the impact of explosive outbursts on family functioning, even when outburst intensity is more resistant to change.

Finally, our findings align with parents' reported need for targeted interventions to help regulate aggression in children with TS [22]. By demonstrating that explosive outbursts are associated with impairments across distinct dimensions of family functioning, the present study supports the clinical relevance of interventions that specifically address explosive outbursts and their functional impact on family life. Interventions that combine strategies to manage outbursts with support aimed at strengthening family problem solving, role distribution, and behavior control may be particularly well suited to address families' needs in this population.

Strengths and Limitations

This study includes a relatively large and international sample, which strengthens the robustness and generalizability

of the findings across diverse family contexts. The use of a validated measure of explosive outburst severity (RAQ-R) is an additional strength, as few studies have assessed explosive outbursts using a standardized severity measure. The focus on the specific impact of explosive outbursts across distinct dimensions of family functioning provides findings with direct clinical implications for interventions targeting explosive outbursts in young people with TS.

Despite this, data were collected through an online survey, which may have increased attrition during questionnaire completion, relied on subjective interpretations of self-report items, and introduced selection biases. Given the focus of this study, parents who chose to participate may also have been those whose children have a more complex presentation of TS. It is therefore possible that the sample may not reflect the experiences of the average parent of a child with TS. In addition, comorbid conditions were reported by parents in terms of their presence rather than severity in order to reduce the length of the survey, which may overlook their potential impact on family functioning.

Future Directions

Future research could use longitudinal designs to examine how changes in explosive outburst severity over time relate to variations in specific dimensions of family functioning, informing causal relationships between these factors. Moreover, incorporating the perspectives of children and adolescents themselves would provide a more comprehensive understanding of explosive outbursts and family functioning, as their perceptions may differ from parent reports and capture aspects not fully reflected in parental ratings.

Conclusion

The present study demonstrates that greater explosive outburst severity in children with TS predicts impairments in general family functioning, problem solving, role distribution, and behavior control. These findings suggest that explosive outbursts have a broad impact on family life, while also affecting core organizational processes and the family's ability to maintain consistent behavioral standards across situations. In contrast, explosive outburst severity did not significantly predict impairments in affective responsiveness, affective involvement, and communication. These findings suggest that the episodic and non-deliberate nature of these outbursts, driven by loss of control rather than intentional aggression, may limit their impact on affective dimensions, and that families may preserve typical interaction patterns outside of outburst episodes. Taken together, these results highlight the unique impact of explosive

outbursts on family functioning, beyond tic severity and comorbid conditions, and underscore the importance of targeting specific family processes in interventions for families of young people with TS.

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Data Availability The data presented in this study are available on request from the corresponding author, upon reasonable request.

Declarations

Ethical Approval This study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by the Ethics Committee of the Université du Québec à Montréal (No. 2024–5786).

Informed Consent Informed consent was obtained from all participants included in the study.

Competing Interests The authors declare no competing interests.

References

1. American Psychiatric Association (2022) Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision (DSM-5-TR). American Psychiatric Association Publishing. <https://doi.org/10.1176/appi.books.9780890425787>
2. Ashurova M, Budman C, Coffey BJ (2021) Ticked Off: Anger Outbursts and Aggressive Symptoms in Tourette Disorder. *Child Adolesc Psychiatr Clin N Am* 30(2):361–373. <https://doi.org/10.1016/j.chc.2020.10.006>
3. Bitsko RH, Holbrook JR, Visser SN, Mink JW, Zinner SH, Ghandour RM, Blumberg SJ (2014) A national profile of Tourette syndrome, 2011–2012. *J Dev Behav Pediatr* 35(5):317–322 <https://doi.org/10.1097/DBP.0000000000000065>
4. Favez N (2010) Chapitre 6. Le modèle du fonctionnement familial de McMaster. In: *L'examen clinique de la famille : Modèles et instruments d'évaluation*. Mardaga, p. 193–214
5. Budman CL, Bruun RD, Park KS, Lesser M, Olson M (2000) Explosive outbursts in children with Tourette's disorder. *J Am Acad Child Adolesc Psychiatry* 39(10):1270–1276 <https://doi.org/10.1097/00004583-200010000-00014>
6. Budman CL, Rockmore L, Stokes J, Sossin M (2003) Clinical phenomenology of episodic rage in children with Tourette syndrome. *J Psychosom Res* 55(1):59–65 [https://doi.org/10.1016/s022-3999\(02\)00584-6](https://doi.org/10.1016/s022-3999(02)00584-6)
7. Budman CL, Rosen M, Shad S (2015) Fits, Tantrums, and Rages in TS and Related Disorders. *Curr Dev Disorders Rep* 2(4):273–284. <https://doi.org/10.1007/s40474-015-0059-1>
8. Cavanna AE (2025) Quality of Life in Tourette Syndrome. *Psychiatr Clin North Am* 48(1):123–150. <https://doi.org/10.1016/j.psyc.2024.08.010>
9. Chang S, Himle MB, Tucker BTP, Woods DW, Piacentini J (2009) Initial Psychometric Properties of a Brief Parent-Report Instrument for Assessing Tic Severity in Children with Chronic Tic Disorders. *Child Fam Behav Ther* 31(3):181–191. <https://doi.org/10.1080/07317100903099100>
10. Chen K, Budman CL, Herrera D, Witkin L, Weiss JE, Lowe NT, Freimer TL, Reus NB, V. I., Mathews CA (2013) Prevalence and clinical correlates of explosive outbursts in Tourette Syndrome. *Psychiatry Res* 205(3):269–275. <https://doi.org/10.1016/j.psychres.2012.09.029>
11. Conelea CA, Woods DW, Zinner SH, Budman C, Murphy T, Scallion LD, Compton SN, Walkup J (2011) Exploring the Impact of Chronic Tic Disorders on Youth: Results from the Tourette Syndrome Impact Survey. *Child Psychiatry Hum Dev* 42(2):219–242. <https://doi.org/10.1007/s10578-010-0211-4>
12. Conte G, Valente F, Fioriello F, Cardona F (2020) Rage attacks in Tourette Syndrome and Chronic Tic Disorder: A systematic review. *Neurosci Biobehavioral Reviews* 119:21–36. <https://doi.org/10.1016/j.neubiorev.2020.09.019>
13. Cox JH, Nahar A, Termine C, Agosti M, Balottin U, Seri S, Cavanna AE (2019) Social stigma and self-perception in adolescents with tourette syndrome. *Adolesc Health Med Ther* 10:75–82. <https://doi.org/10.2147/AHMT.S175765>
14. Cravedi E, Deniau E, Giannitelli M, Xavier J, Hartmann A, Cohen D (2017) Tourette syndrome and other neurodevelopmental disorders: A comprehensive review. *Child Adolesc Psychiatry Mental Health* 11(1):59. <https://doi.org/10.1186/s13034-017-0196-x>
15. Dai L, Wang L (2015) Review of Family Functioning. *Open J Social Sci* 03(12):134–141. <https://doi.org/10.4236/jss.2015.312014>
16. De Lange N, Olivier M (2004) Mothers' Experiences of Aggression in Their Tourette's Syndrome Children. *Int J Advancement Counselling* 26(1):65–77. <https://doi.org/10.1023/B:ADCO.0000021550.87868.bd>
17. Dooley JM, Brna PM, Gordon KE (1999) Parent perceptions of symptom severity in Tourette's syndrome. *Arch Dis Child* 81:440–441 <https://doi.org/10.1136/adc.81.5.440>
18. Eapen V, Cavanna AE, Robertson MM (2016) Comorbidities, Social Impact, and Quality of Life in Tourette Syndrome. *Front Psychiatry* 7:97 <https://doi.org/10.3389/fpsy.2016.00097>
19. Eddy CM, Rizzo R, Gulisano M, Agodi A, Barchitta M, Cali P, Robertson MM, Cavanna AE (2011) Quality of life in young people with Tourette syndrome: A controlled study. *J Neurol* 258(2):291–301. <https://doi.org/10.1007/s00415-010-5754-6>
20. Epstein NB, Baldwin LM, Bishop DS (1983) The McMaster Family Assessment Device. *J Marital Fam Ther* 9(2):171–180. <https://doi.org/10.1111/j.1752-0606.1983.tb01497.x>

21. Epstein NB, Bishop DS, Levin S (1978) The McMaster Model of Family Functioning. *J Marital Fam Ther* 4(4):19–31. <https://doi.org/10.1111/j.1752-0606.1978.tb00537.x>
22. Evans G, Wittkowski A, Butler H, Hedderly T, Bunton P (2015) Parenting interventions in tic disorders: An exploration of parents' perspectives. *Child Care Health Dev* 41(3):384–396. <https://doi.org/10.1111/cch.12212>
23. Ferreira BR, Pio-Abreu JL, Januário C (2014) Tourette's syndrome and associated disorders: A systematic review. *Trends Psychiatry Psychother* 36(3):123–133. <https://doi.org/10.1590/2237-6089-2014-1003>
24. Ghanizadeh A, Mohammadi M-R, Dehbozorgi GR (2010) Children and Their Parent's Perceptions of Symptom Severity and Treatment Preference for Tourette Syndrome. *Iran J Psychiatry* 5(3):93–96
25. Greenberg E, Lazaroaie F, Budman C (2026) Explosive outbursts in Tourette syndrome and chronic tic disorders. Dans *Handbook of Clinical Neurology* (Vol. 215, pp. 97–109). Elsevier. <https://doi.org/10.1016/B978-0-443-13554-5.00014-6>
26. Groth C (2018) Tourette syndrome in a longitudinal perspective. Clinical course of tics and comorbidities, coexisting psychopathologies, phenotypes and predictors. *Dan Med J* 65(4):B5465
27. Hirschtritt ME, Lee PC, Pauls DL, Dion Y, Grados MA, Illmann C, King RA, Sandor P, McMahon WM, Lyon GJ, Cath DC, Kurlan R, Robertson MM, Osiecki L, Scharf JM, Mathews CA (2015) Lifetime Prevalence, Age of Risk, and Genetic Relationships of Comorbid Psychiatric Disorders in Tourette Syndrome. *JAMA Psychiatry* 72(4):325. <https://doi.org/10.1001/jamapsychiatry.2014.2650>
28. Johnson KA, Worbe Y, Foote KD, Butson CR, Gunduz A, Okun MS (2023) Tourette syndrome: Clinical features, pathophysiology, and treatment. *Lancet Neurol* 22(2):147–158. [https://doi.org/10.1016/S1474-4422\(22\)00303-9](https://doi.org/10.1016/S1474-4422(22)00303-9)
29. Johnston C, Ohan JL (2005) The Importance of Parental Attributions in Families of Children with Attention-Deficit/Hyperactivity and Disruptive Behavior Disorders. *Clin Child Fam Psychol Rev* 8(3):167–182. <https://doi.org/10.1007/s10567-005-6663-6>
30. Knight T, Steeves T, Day L, Lowerison M, Jette N, Pringsheim T (2012) Prevalence of Tic Disorders: A Systematic Review and Meta-Analysis. *Pediatr Neurol* 47(2):77–90. <https://doi.org/10.1016/j.pediatrneurol.2012.05.002>
31. Lecavalier L, Bodfish J, Harrop C, Whitten A, Jones D, Pritchett J, Faldowski R, Boyd B (2020) Development of the Behavioral Inflexibility Scale for Children with Autism Spectrum Disorder and Other Developmental Disabilities. *Autism Res* 13(3):489–499. <https://doi.org/10.1002/aur.2257>
32. Leclerc J, Forget J, O'Connor KP (2008) Quand le corps fait à sa tête: Le syndrome de Gilles de la Tourette. Éditions MultiMondes
33. Leclerc J, O'Connor KP, Forget J, Lavoie ME (2012) Évaluation de l'effet d'un programme d'entraînement à l'autogestion des épisodes explosifs chez des enfants atteints du syndrome de Gilles de la Tourette. *Pratiques Psychologiques* 18(3):221–244. <https://doi.org/10.1016/j.prps.2010.07.001>
34. Liu S, Zheng L, Zheng X, Zhang X, Yi M, Ma X (2017) The Subjective Quality of Life in Young People With Tourette Syndrome in China. *J Atten Disord* 21(5):426–432. <https://doi.org/10.1177/1087054713518822>
35. Mansfield AK, Keitner GI, Dealy J (2015) The Family Assessment Device: An Update. *Fam Process* 54(1):82–93. <https://doi.org/10.1111/famp.12080>
36. M.Blanchet Mathieu (2021) Les épisodes explosifs chez les enfants ayant un syndrome de Gilles de la Tourette: Effet d'une thérapie sur le fonctionnement familial et exploration de l'incidence des troubles associés sur l'efficacité du traitement. Université du Québec à Montréal <https://archipel.uqam.ca/14705/1/D3962.pdf>
37. McGuire JF, Piacentini J, Storch EA, Ricketts EJ, Woods DW, Peterson AL, Walkup JT, Wilhelm S, Ramsey K, Essoe JK-Y, Himle MB, Lewin AB, Chang S, Murphy TK, McCracken JT, Scahill L (2021) Defining tic severity and tic impairment in Tourette Disorder. *J Psychiatr Res* 133:93–100. <https://doi.org/10.1016/j.jpsychires.2020.12.040>
38. Miller IW, Ryan CE, Keitner GI, Bishop DS, Epstein NB (2000) The McMaster Approach to Families: Theory, assessment, treatment and research. *J Family Therapy* 22(2):168–189. <https://doi.org/10.1111/1467-6427.00145>
39. Morand-Beaulieu S, Gagnon M, Leclerc JB (2026). The role of behavioral inflexibility and impulsivity in explosive outbursts among youth with Tourette syndrome. *Eur Child Adolesc Psychiatry* <https://doi.org/10.1007/s00787-026-03176-3>
40. Morand-Beaulieu S, Grot S, Lavoie J, Leclerc JB, Luck D, Lavoie ME (2017) The puzzling question of inhibitory control in Tourette syndrome: A meta-analysis. *Neurosci Biobehavioral Reviews* 80:240–262. <https://doi.org/10.1016/j.neubiorev.2017.05.006>
41. Müller-Vahl KR, Kayser L, Pisarenko A, Haas M, Psathakis N, Palm L, Jakubovski E (2020) The Rage Attack Questionnaire-Revised (RAQ-R): Assessing Rage Attacks in Adults With Tourette Syndrome. *Front Psychiatry* 10:956. <https://doi.org/10.3389/fpsy.2019.00956>
42. Pappert EJ, Goetz CG, Louis ED, Blasucci L, Leurgans S (2003) Objective assessments of longitudinal outcome in Gilles de la Tourette's syndrome. *Neurology* 61(7):936–940. <https://doi.org/10.1212/01.WNL.0000086370.10186.7C>
43. Quast LF, Rosenthal LD, Cushman GK, Gutiérrez-Colina AM, Braley EI, Kardon P, Blount RL (2020) Relations Between Tic Severity, Emotion Regulation, and Social Outcomes in Youth with Tourette Syndrome. *Child Psychiatry Hum Dev* 51(3):366–376. <https://doi.org/10.1007/s10578-019-00948-8>
44. Ricketts EJ, McGuire JF, Chang S, Bose D, Rasch MM, Woods DW, Specht MW, Walkup JT, Scahill L, Wilhelm S, Peterson AL, Piacentini J (2018) Benchmarking Treatment Response in Tourette's Disorder: A Psychometric Evaluation and Signal Detection Analysis of the Parent Tic Questionnaire. *Behav Ther* 49(1):46–56. <https://doi.org/10.1016/j.beth.2017.05.006>
45. Rivera-Navarro J, Cubo E, Almazán J (2014) The Impact of Tourette's Syndrome in the School and the Family: Perspectives from Three Stakeholder Groups. *Int J Advancement Counselling* 36(1):96–113. <https://doi.org/10.1007/s10447-013-9193-9>
46. Robinson LR, Bitsko RH, Schieve LA, Visser SN (2013) Tourette syndrome, parenting aggravation, and the contribution of co-occurring conditions among a nationally representative sample. *Disabil Health J* 6(1):26–35. <https://doi.org/10.1016/j.dhjo.2012.10.002>
47. Sambrani T, Jakubovski E, Müller-Vahl KR (2016) New Insights into Clinical Characteristics of Gilles de la Tourette Syndrome: Findings in 1032 Patients from a Single German Center. *Front Neurosci* 10. <https://doi.org/10.3389/fnins.2016.00415>
48. Scahill L, Specht M, Page C (2014) The prevalence of tic disorders and clinical characteristics in children. *J Obsessive-Compulsive Relat Disorders* 3(4):394–400. <https://doi.org/10.1016/j.jocrd.2014.06.002>
49. Scharf JM, Miller LL, Gauvin CA, Alabiso J, Mathews CA, Ben-Shlomo Y (2015) Population prevalence of Tourette syndrome: A systematic review and meta-analysis. *Mov Disord* 30(2):221–228. <https://doi.org/10.1002/mds.26089>
50. Staccini L, Tomba E, Grandi S, Keitner GI (2015) The Evaluation of Family Functioning by the Family Assessment Device: A Systematic Review of Studies in Adult Clinical Populations. *Fam Process* 54(1):94–115. <https://doi.org/10.1111/famp.12098>
51. Storch EA, Johnco C, McGuire JF, Wu MS, McBride NM, Lewin AB, Murphy TK (2017) An initial study of family accommodation in children and adolescents with chronic tic disorders. *Eur*

- Child Adolesc Psychiatry 26(1):99–109. <https://doi.org/10.1007/s00787-016-0879-5>
52. Sukhodolsky DG, Scahill L, Zhang H, Peterson BS, King RA, Lombroso PJ, Katsoyich L, Findley D, Leckman JF (2003) Disruptive Behavior in Children With Tourette's Syndrome: Association With ADHD Comorbidity, Tic Severity, and Functional Impairment. *J Am Acad Child Adolesc Psychiatry* 42(1):98–105. <https://doi.org/10.1097/00004583-200301000-00016>
 53. Wilkinson BJ, Newman MB, Shytte RD, Silver AA, Sanberg PR, Sheehan D (2001) Family Impact of Tourette's Syndrome. *J Child Fam stud* 10(4):477–483. <https://doi.org/10.1023/A:1016713508665>
 54. Woods DW, Himle MB, Osmon DC (2005) Use of the Impact on Family Scale in Children with Tic Disorders: Descriptive Data, Validity, and Tic Severity Impact. *Child Fam Behav Ther* 27(2):11–21. https://doi.org/10.1300/J019v27n02_02
 55. Yang J, Hirsch L, Martino D, Jette N, Roberts J, Pringsheim T (2016) The prevalence of diagnosed tourette syndrome in Canada: A national population-based study. *Mov Disord* 31(11):1658–1663. <https://doi.org/10.1002/mds.26766>

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